

FATIMA JINNAH WOMEN UNIVERSITY

Department of Software Engineering
Human Computer Interaction Lab
Semester Project Report | Section VI-B

Fighter Jet Shooting Game

In C#

SUBMITTED BY

Manahil Habib (2023-BSE-042)
Maria Irfan (2023-BSE-044)

SUBMITTED TO

Sir Rehan

1. Introduction

The “**Fighter Jet Shooter Game**” is a simple 2D shooting game developed as a semester project for the HCI Lab course. The purpose of this project was to design an interactive and user-friendly game interface while implementing basic gaming concepts such as movement, shooting, enemy interaction, collision handling and score tracking.

The game contains one player fighter jet controlled by the user and three enemy fighter jets placed at the top side of the screen. The player shoots bullets toward enemies and gains score during gameplay. The project combines graphical designing and programming concepts to create an engaging gaming experience.

2. Objective of the Project

The main objectives of this project are:

- To understand Human Computer Interaction concepts.
- To design an interactive graphical user interface.
- To implement event-driven programming in C#.
- To create a simple shooting game using Windows Forms.
- To improve user experience through visuals and controls.

3. Tools and Technologies Used

Tool / Technology	Purpose
Adobe Illustrator	Designing game visuals and graphics
Visual Studio	Coding and development
C# Language	Game logic implementation
Windows Forms	GUI development
PictureBox Controls	Displaying jets, bullets, and clouds
Timer Component	Continuous game movement

4. Game Design and Development Process

The Fighter Jet Shooter Game was first designed in Adobe Illustrator. Different graphical assets such as fighter jets, clouds and bullets were created to make the interface visually attractive. After designing the assets, they were imported into Visual Studio and used inside the Windows Forms application through the Resources folder.

The game was developed using C# and event-driven programming concepts. PictureBox controls were used to display all visual game objects including the player jet, enemy jets, bullets and mov-

ing clouds.

A Timer component continuously updates the gameplay mechanics such as:

- Enemy movement
- Bullet movement
- Collision detection
- Cloud animation
- Score updates

The game environment was designed with a sky-blue background to simulate an air combat scene.

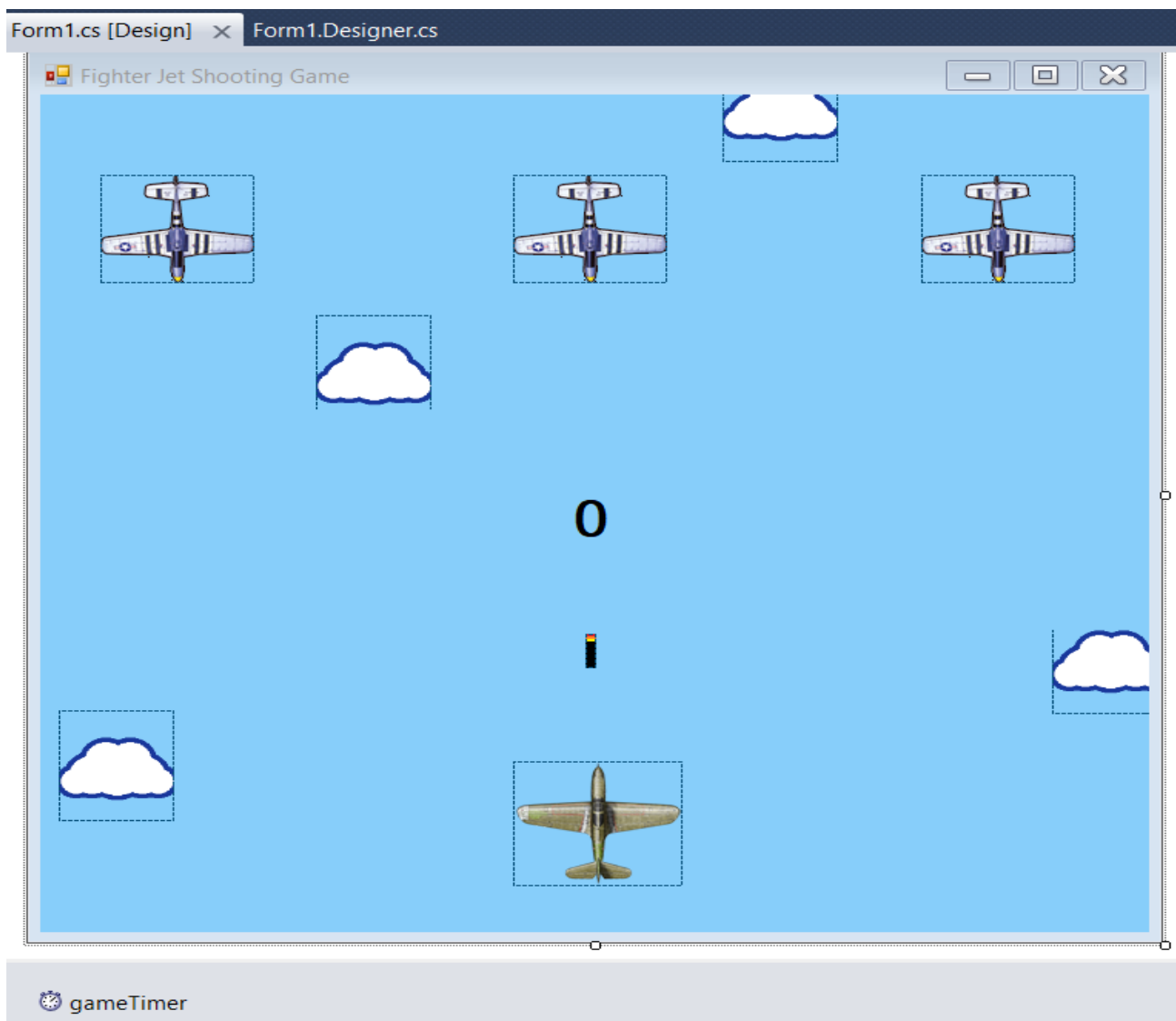


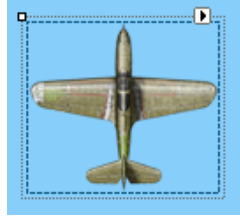
Figure 1 Main interface of the Fighter Jet Shooter Game developed in C# Windows Forms

5. Game Features

The Fighter Jet Shooter Game includes the following features:

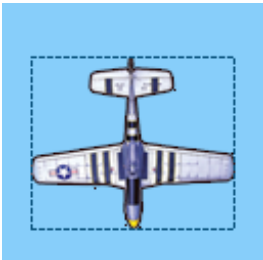
Player Fighter Jet

- Controlled using keyboard arrow keys.
- Can move left and right.
- Fires bullets using the Space key.



Enemy Jets

- Three enemy fighter jets appear from the top.
- Enemy speed increases as the score increases.
- Enemies respawn at random positions after being hit.



Bullet Shooting System

- Only one bullet is fired at a time.
- Bullet travels upward toward enemy jets.
- Collision detection increases the player score.

Dynamic Difficulty

Enemy speed changes according to the score:

Score Enemy Speed

0	5
5	7
10	9
15	11

This creates a progressively challenging gameplay experience.

6. Cloud Animation System

A cloud animation system was implemented to improve the visual appearance of the game.

Features of Cloud System:

- Multiple clouds move downward continuously.
- Clouds reset automatically after leaving the screen.
- Each cloud has a different movement speed, Random positions create a natural sky effect.

This feature gives a depth-like effect and improves the user experience.

7. Working of the Game

When the game starts the player controls a fighter jet located at the bottom of the screen. The player uses the Left and Right arrow keys to move the jet.

When the Space key is pressed:

- A bullet is fired upward.
- The bullet moves toward enemy jets.
- If the bullet collides with an enemy jet:
 - The score increases.
 - The enemy respawns at a random location.

Enemy jets continuously move downward. If any enemy crosses the bottom boundary of the screen, the game ends and a “Game Over” message is displayed.

The player can restart the game by pressing the Enter key.

8. Keyboard Controls

Key	Function
Left Arrow	Move player left
Right Arrow	Move player right
Space Bar	Shoot bullet
Enter	Restart game

9. Important Programming Concepts Used

Event-Driven Programming

Keyboard and timer events control all game interactions.

Collision Detection

The `IntersectsWith()` method is used to detect bullet collisions with enemy jets.

Example:

```
if (bullet.Bounds.IntersectsWith(enemyThree.Bounds))  
  
    {  
        score += 1;  
    }
```

Randomization

Random positions are generated for enemy jets and clouds using the `Random` class.

Example:

```
enemyOne.Left = rnd.Next(20, 450);
```

Timer-Based Animation

The `Timer` component updates gameplay continuously every 20 milliseconds.

Example:

```
this.gameTimer.Interval = 20;
```

Object-Oriented Programming

A separate `Cloud` class was created to manage cloud movement and behavior.

This improved:

- Code organization
- Reusability
- Readability

10. HCI Concepts Applied

The following Human Computer Interaction principles were implemented in this project:

Simplicity

The interface is simple and easy for users to understand.

User Control

Users directly control the fighter jet using keyboard inputs.

Visual Consistency

All graphical elements follow a consistent aviation theme.

Immediate Feedback

The score updates instantly when enemies are hit.

Interactive Environment

Moving clouds and enemy jets create an engaging user experience.

11. Challenges Faced During Development

Several challenges were faced during the development process:

- Designing graphics in Adobe Illustrator.
- Importing graphical assets into Visual Studio.
- Implementing smooth movement using timer events.
- Handling bullet collision detection.
- Managing cloud animation and random movement.
- Increasing enemy difficulty dynamically.
- Synchronizing user controls with gameplay mechanics.

12. Future Improvements

The following features can be added in future versions:

- Multiple bullets firing system
- Enemy shooting mechanism
- Sound effects and background music
- Explosion animations
- Multiple game levels
- Health and life system
- Pause and resume option
- Leaderboard and high scores
- More advanced enemy AI

13. Conclusion

The Fighter Jet Shooter Game is a 2D shooting game developed using C# Windows Forms and Adobe Illustrator. The project successfully combines graphical designing with programming concepts to create an interactive gaming environment.

The game demonstrates important HCI principles such as user interaction, visual feedback, consistency and simplicity. Features such as moving clouds, collision detection, dynamic enemy speed and keyboard controls improve the overall gameplay experience.

This project helped in understanding:

- GUI development
- Event-driven programming
- Basic game mechanics
- Human Computer Interaction concepts

The project also enhanced practical skills in both software development and graphical interface design.
